Teacher answer key: Primary 3, Semester 2 (DRAFT)

DRAFT: this answer key is for owner review. Do not give it to pupils. Contact: pypteltd@gmail.com (business email for now, to be changed later)

This key lists the answers and the result of the check on every page. You do not need it for routine marking: a pupil who has a working check has the right answers. To spot-check, compare the check strip with the line "Check result" below.

How the topic tags work

Each entry names the syllabus outcomes the puzzle practises, with the outcome numbers of the Primary 3 content in the Mathematics Syllabus, Primary (2021 edition, pages 35 and 36). The references are for your planning only and are not printed on the pupil pages. "Problem solving" marks a page whose questions come from the kinds found in school tests rather than from one numbered outcome.

Pages in this booklet (Terms 3 and 4)

Strand	Pages
Measurement and Geometry - Measurement	S2-01, S2-02, S2-03, S2-04, S2-05, S2-06, S2-07, S2-08, S2-09, S2-19
Measurement and Geometry - Area and Perimeter	S2-10, S2-11, S2-12, S2-13
Measurement and Geometry - Geometry	S2-14, S2-15, S2-16
Statistics - Bar Graphs	S2-17, S2-18
Number and Algebra - Money	S2-20

The page code in the footer, for example S2-07 (P3-27), gives the page in this booklet and, in brackets, its number in the full set of 40.

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S2-01 (P3-21) Jugs and Beakers 1 star

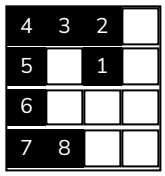
Topic: Volume of liquid: reading containers in millilitres (mℓ)

Syllabus outcomes: Length, Mass and Volume 1.1. Notes on the tag: 1.1 (measuring volume of liquid in millilitres)
Puzzle type: Code-breaker (containers). How the pupil checks: Code word (6 letters).
Answers: 1) 100 mℓ (scale to 1000, a mark every 100); 2) 700 mℓ (scale to 1000, a mark every 100); 3) 300 mℓ (scale to 1000, a mark every 100); 4) 450 mℓ (scale to 500, a mark every 50); 5) 150 mℓ (scale to 500, a mark every 50); 6) 500 mℓ (scale to 1000, a mark every 100)
Check result: Code word: DONKEY
Teacher note: Six containers with a scale in millilitres: some to 500 mℓ with a mark every 50 mℓ, some to 1000 mℓ with a mark every 100 mℓ, and numbers on only some of the marks, so the pupil counts on from a number. The level is at a mark, never at a number. The decoy letters stand for the mark above or below. Drawings are not to scale. Practical measuring with a real jug is not possible on a printed page. A box left blank gives no letter and the check cannot see it.

S2-02 (P3-22) Equal Measures 1 star

Topic: Converting between km and m, m and cm, kg and g, ℓ and mℓ

Syllabus outcomes: Length, Mass and Volume 1.2, Length, Mass and Volume 1.3. Notes on the tag: 1.2, 1.3 (compound units; converting km and m, m and cm, kg and g, ℓ and mℓ)
Puzzle type: Maze / path. How the pupil checks: Route: the line must reach FINISH through the right number of boxes, every missing number being 4000.
Answers: Route (row, column from top-left): (2,3) 3 ℓ + 1000 mℓ = ? mℓ -> (1,3) 1 km + 3000 m = ? m -> (1,2) 4 kg = ? g -> (1,1) 3 kg + 1000 g = ? g -> (2,1) 2 ℓ + 2000 mℓ = ? mℓ -> (3,1) 2 km + 2000 m = ? m -> (4,1) 4 km = ? m -> (4,2) 3 km + 1000 m = ? m



Check result: Route reaches FINISH through 8 boxes; the missing number in every box on it is 4000.
Teacher note: A 4 by 4 grid of sixteen conversions with a missing number: km to m, kg to g, ℓ to mℓ, m to cm, and sums of a larger and a smaller unit that must be written in the smaller unit. The route follows the boxes whose missing number is 4000. The other boxes have missing numbers that are a thousand more or less, or ten or a hundred times as big or as small. 2 extra boxes that do make 4000 are dead ends. Counting the boxes shows a line that takes a shortcut. Pictures of rulers, scales and jugs are not used: the page tests the conversions.

S2-03 (P3-23) Kilograms and Grams 1 star

Topic: Mass: reading scales and changing kg to g

Syllabus outcomes: Length, Mass and Volume 1.1, Length, Mass and Volume 1.3. Notes on the tag: 1.1, 1.3 (measuring mass; converting kg and g)
Puzzle type: Code-breaker (weighing scales). How the pupil checks: Code word (6 letters).
Answers: 1) 650 g = 650 g; 2) 950 g = 950 g; 3) 9 kg = 9000 g; 4) 4 kg = 4000 g; 5) 8 kg = 8000 g; 6) 850 g = 850 g
Check result: Code word: FRUITY
Teacher note: Six weighing scales with a needle: three read in grams (0 to 1000 g, a mark halfway between the numbers, the readings are on such a mark) and three read in kilograms (0 to 10 kg) that the pupil changes to grams. Every answer is written in grams. The decoy letters stand for the next mark read by mistake, or for the number of kilograms written as grams without the zeros. Drawings are not to scale. A box left blank gives no letter and the check cannot see it.

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S2-04 (P3-24) Road Trip 2 stars

Topic: Compound units: adding and subtracting km and m, kg and g, ℓ and mℓ

Syllabus outcomes: Length, Mass and Volume 1.1, Length, Mass and Volume 1.2, Length, Mass and Volume 1.3. Notes on the tag: 1.1, 1.2 (measuring length in kilometres and volume in millilitres; compound units)
Puzzle type: Code-breaker (compound units). How the pupil checks: Code word (6 letters).
Answers: 1) 4500 m; 2) 4700 m; 3) 7350 g; 4) 2500 g; 5) 4000 mℓ; 6) 4100 mℓ
Check result: Code word: WALNUT
Teacher note: Six word problems with compound units: two about length (km and m), two about mass (kg and g), two about volume (ℓ and mℓ). Four add or subtract two compound measures and two take a part away. Every answer is written in the smaller unit. Subtractions need the larger unit to be renamed. The decoy letters stand for forgetting to rename, or for the wrong number of thousands. A box left blank gives no letter and the check cannot see it.

S2-05 (P3-25) Order the Measures 2 stars

Topic: Length, mass and volume: comparing and ordering measures in different units

Syllabus outcomes: Length, Mass and Volume 1.2, Length, Mass and Volume 1.3. Notes on the tag: 1.2, 1.3 (compound units; converting km and m, kg and g, ℓ and mℓ)
Puzzle type: Ordering (lettered cards). How the pupil checks: Letters of the cards in the order asked spell a word (6 letters).
Answers: Set A (shortest first): W = 2 km 5 m, O = 2 km 50 m, M = 2500 m; Set B (greatest first): B = 7400 g, A = 7 kg 40 g, T = 7 kg 4 g
Check result: Word: WOMBAT
Teacher note: Each set has three measures of the same kind written in different ways (a mixed unit such as km and m, or only the smaller unit). To compare them the pupil changes them all to the smaller unit. The traps are the zeros in the middle (50 m, 5 m) and the number of thousands. The letters, read in the order asked, spell a word with exactly one arrangement that is a word, so no wrong order of the six cards spells a word (every wrong order was tested against a very large word list). A box left blank gives no letter and the check cannot see it.

S2-06 (P3-26) Seconds 1 star

Topic: Time: seconds, minutes and seconds

Syllabus outcomes: Time 2.1. Notes on the tag: 2.1 (measuring time in seconds)
Puzzle type: Code-breaker (stopwatch). How the pupil checks: Code word (6 letters).
Answers: 1) 30 s; 2) 95 s; 3) 300 s; 4) 280 s; 5) 6 min; 6) 70 s
Check result: Code word: DEVOUR
Teacher note: Two stopwatch faces (a seconds hand, a mark every second and a number every 5 seconds; the second also says that one minute has passed), a minutes-and-seconds measure changed to seconds, the number of seconds in some minutes, a story, and seconds changed to minutes. The decoy letters stand for reading the wrong mark, writing a minute as a hundred seconds, or adding the minutes to the seconds. A real stopwatch is not possible on paper. A box left blank gives no letter and the check cannot see it.

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S2-07 (P3-27) The 24-Hour Clock 1 star

Topic: Time: the 24-hour clock

Syllabus outcomes: Time 2.3. Notes on the tag: 2.3 (24-hour clock)
Puzzle type: Lettered choice (24-hour clock). How the pupil checks: Word from the letters of the right choices (5 letters).
Answers: 1) 3:05 am = 0305 h; 2) 9:36 am = 0936 h; 3) 2:05 pm = 1405 h; 4) 10:12 pm = 2212 h; 5) 6:30 pm = 1830 h
Check result: Word: SAUCY
Teacher note: Five rows: two analogue clocks (with the part of the day written beside them), two times in the 12-hour clock to write in the 24-hour clock and one 24-hour time to write in the 12-hour clock. In the 24-hour clock the day is counted from midnight and every time is written with four digits and the letter h. The wrong choices are the same time in the other half of the day, the hour and the minutes swapped, or a time ten minutes or two hours away. The letters of the right choices, read down the rows, spell the word. Every other way of choosing was tested against a very large word list: none spells a word. A box left blank gives no letter and the check cannot see it.

S2-08 (P3-28) Start and Finish 2 stars

Topic: Time: finding the starting time, the finishing time or how long it takes

Syllabus outcomes: Time 2.2. Notes on the tag: 2.2 (finding the starting time, finishing time or duration given the other two quantities)
Puzzle type: Lettered choice (start, finish, duration). How the pupil checks: Word from the letters of the right choices (6 letters).
Answers: 1) 2:25 pm; 2) 55 min; 3) 6:15 am; 4) 2:10 pm; 5) 1 h 50 min; 6) 5:25 pm
Check result: Word: VOLUME
Teacher note: Six stories: two ask when something ends (start and how long), two ask when it started (end and how long) and two ask how long it took (start and end). Times are written with am and pm; some go past the hour, some go past noon. The wrong choices are the usual slips: counting a hundred minutes in an hour, adding where the story takes away, and a time out by an hour or by forty minutes. The letters of the right choices, read down the rows, spell the word. Every other way of choosing was tested against a very large word list: none spells a word. A box left blank gives no letter and the check cannot see it.

S2-09 (P3-29) Time Detective 2 stars

Topic: Time: the 24-hour clock, start, finish and how long

Syllabus outcomes: Time 2.2, Time 2.3. Notes on the tag: 2.2, 2.3 (starting time, finishing time or duration; 24-hour clock)
Puzzle type: Code-breaker (timetable). How the pupil checks: Code word (6 letters).
Answers: 1) 85 min; 2) 80 min; 3) 1635 h; 4) 0805 h; 5) 150 min; 6) 55 min
Check result: Code word: WISDOM
Teacher note: A timetable of buses in the 24-hour clock (four digits, with h). Questions: how long three journeys take (one crosses midnight), when a journey that leaves at a given time and takes a given time arrives, when a journey that arrives at a given time and takes a given time left, and the wait between two buses. Answers that are times are written with four digits (a leading zero for the early hours); answers that are lengths are in minutes. The decoy letters stand for counting a hundred minutes in an hour, or for an hour out. A box left blank gives no letter and the check cannot see it.

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S2-10 (P3-30) Square Units 1 star

Topic: Area and perimeter: counting square units

Syllabus outcomes: Area and Perimeter 1.1, Area and Perimeter 1.2. Notes on the tag: 1.1, 1.2 (concepts of area and perimeter; area in square units and cm2)
Puzzle type: Code-breaker (area and perimeter). How the pupil checks: Code word (6 letters).
Answers: Figure A: area 8 cm2, perimeter 14 cm; Figure B: area 9 cm2, perimeter 16 cm; Figure C: area 12 cm2, perimeter 18 cm
Check result: Code word: CAMPUS
Teacher note: Three figures made of 1 cm squares (every square is 8 mm on the page, so the figures are drawn smaller than real size). For each figure the pupil counts the squares (area) and the units round the edge (perimeter), so that the two ideas are met side by side. The decoy letters stand for the area and the perimeter swapped, or for one square or one unit out. No figure has a hole. Area and perimeter of rectilinear figures given on a grid are Primary 3; missing sides are not asked. A box left blank gives no letter and the check cannot see it.

S2-11 (P3-31) Perimeter 1 star

Topic: Perimeter of rectangles, squares and rectilinear figures

Syllabus outcomes: Area and Perimeter 1.3, Area and Perimeter 1.1. Notes on the tag: 1.3 (perimeter of a rectilinear figure, a rectangle and a square)
Puzzle type: Code-breaker (perimeter). How the pupil checks: Code word (6 letters).
Answers: 1) 32; 2) 16; 3) 150; 4) 36; 5) 42; 6) 38
Check result: Code word: PEANUT
Teacher note: Six items: a rectangle (two sides given), a square (one side given), three rectilinear figures with every side labelled (an L shape, a figure with a notch and a staircase) and a field in a story. Figures are drawn to scale but pictures are not actual size. The decoy letters stand for adding only the labelled sides of a rectangle once, or for half the perimeter. A box left blank gives no letter and the check cannot see it.

S2-12 (P3-32) Area Sums 2 stars

Topic: Area of rectangles and squares in cm² and m²

Syllabus outcomes: Area and Perimeter 1.2, Area and Perimeter 1.4. Notes on the tag: 1.2, 1.4 (area in cm2 and m2; area of a rectangle and a square)
Puzzle type: Code-breaker (area). How the pupil checks: Code word (6 letters).
Answers: 1) 40 m2; 2) 49 m2; 3) 36 cm2; 4) 84 m2; 5) 70 cm2; 6) 64 cm2
Check result: Code word: BANDIT
Teacher note: Six areas: a rectangle and a square given in a picture, a garden and a floor given in words (in m²), a square card and a rectangle covered with 1 cm squares (in cm²). The unit is printed after the box, so cm² and m² are never changed into each other. The decoy letters stand for the perimeter, or for adding the two sides. A box left blank gives no letter and the check cannot see it.

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S2-13 (P3-33) Same or Different? 2 stars

Topic: Area and perimeter: comparing two figures made of unit squares

Syllabus outcomes: Area and Perimeter 1.1, Area and Perimeter 1.2, Area and Perimeter 1.3. Notes on the tag: 1.1, 1.2, 1.3 (concepts of area and perimeter; area in square units; perimeter of a rectilinear figure)
Puzzle type: Lettered choice (compare two figures). How the pupil checks: Word from the letters of the right choices (5 letters).
Answers: 1) B (area); 2) Same (perimeter); 3) B (area); 4) A (perimeter); 5) B (perimeter)
Check result: Word: WORKS
Teacher note: Five pairs of figures made of equal squares (each square is a unit). The pupil counts the squares for the area and the units round the edge for the perimeter, and says which figure is greater or whether they are the same. Pairs are chosen so that the answers include A, B and the same. A figure with more squares need not have the greater perimeter. The letters of the right choices, read down the rows, spell the word. Every other way of choosing was tested against a very large word list: none spells a word. A box left blank gives no letter and the check cannot see it.

S2-14 (P3-34) Right Angles 1 star

Topic: Angles: right angles, smaller and greater than a right angle

Syllabus outcomes: Angles 1.1, Angles 1.2. Notes on the tag: 1.1, 1.2 (concept of an angle; right angles; angles greater than or smaller than a right angle)
Puzzle type: Lettered choice (angles). How the pupil checks: Word from the letters of the right choices (5 letters).
Answers: 1) smaller than a right angle; 2) a right angle; 3) smaller than a right angle; 4) a right angle; 5) greater than a right angle
Check result: Word: BIRCH
Teacher note: Five angles, each drawn with a small arc. Each angle is clearly smaller than a right angle, exactly a right angle or clearly greater: none is close to a right angle, so a pupil can compare with the corner of a sheet of paper or a set square. Degrees are not named on the page: the page asks only for right angles and smaller or greater. The angles are turned to different positions, so a right angle is not always a horizontal and a vertical line. The letters of the right choices, read down the rows, spell the word. Every other way of choosing was tested against a very large word list: none spells a word. A box left blank gives no letter and the check cannot see it.

S2-15 (P3-35) Line Detective 1 star

Topic: Perpendicular and parallel lines: identifying them

Syllabus outcomes: Perpendicular and Parallel Lines 2.1. Notes on the tag: 2.1 (identifying perpendicular and parallel lines)
Puzzle type: Lettered choice (lines). How the pupil checks: Word from the letters of the right choices (5 letters).
Answers: 1) line R (perpendicular); 2) line R (parallel); 3) line Q (parallel); 4) line Q (parallel); 5) line P (perpendicular)
Check result: Word: BLUNT
Teacher note: Five pictures, each with the thick line AB and three thin lines P, Q and R. In every picture one line is parallel to AB, one is perpendicular to AB and one is neither; the question asks for the parallel or the perpendicular one. Lines are drawn from slopes that fit a grid, but no grid is printed. The pupil can use a ruler and the corner of a sheet of paper. Parallel means always the same distance apart; perpendicular means meeting at a right angle (the lines are not made to touch). The letters of the right choices, read down the rows, spell the word. Every other way of choosing was tested against a very large word list: none spells a word. A box left blank gives no letter and the check cannot see it.

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S2-16 (P3-36) Draw the Lines 2 stars

Topic: Perpendicular and parallel lines: drawing them

Syllabus outcomes: Perpendicular and Parallel Lines 2.2. Notes on the tag: 2.2 (drawing perpendicular and parallel lines)
Puzzle type: Draw and read off (lettered edge). How the pupil checks: Word from the letters where the drawn lines end (6 letters).
Answers: 1) perpendicular through P: letter S at x = 3; 2) perpendicular through P: letter W at x = 2; 3) parallel through P: letter I at x = 2; 4) parallel through P: letter V at x = 5; 5) perpendicular through P: letter E at x = 5; 6) parallel through P: letter L at x = 2
Check result: Word: SWIVEL
Teacher note: Six grids with a thick line AB, a point P and five letters on the top edge. The pupil draws a line through P that is perpendicular (or parallel) to AB, as far as the top edge, and writes the letter where it ends. The right letter is on a grid point of the top edge; the letters beside it are where the line ends if the pupil draws the wrong direction. The page cannot see the drawing: it sees the letter, which is the printed evidence of the drawing, and the teacher can look at the lines. Drawing is checked only through the letter. A box left blank gives no letter and the check cannot see it.

S2-17 (P3-37) Bar Graph Detective 1 star

Topic: Bar graphs: reading and interpreting

Syllabus outcomes: Bar Graphs 1.1. Notes on the tag: 1.1 (reading and interpreting bar graphs)
Puzzle type: Code-breaker (bar graph). How the pupil checks: Code word (6 letters).
Answers: 1) 15; 2) 40; 3) 30; 4) 25; 5) 120; 6) 10
Check result: Code word: SPRUCE
Teacher note: A bar graph of five clubs on a scale from 0 to 50 with a line for every 5 pupils and a number for every 10, so most bars end on a line between two numbers. Six questions: reading one bar, the tallest bar, a difference, the sum of two bars, the total of all five bars and the least. The decoy letters stand for reading the next line, or for adding the wrong two bars. The bars are black; no colour is needed. A box left blank gives no letter and the check cannot see it.

S2-18 (P3-38) Different Scales 2 stars

Topic: Bar graphs with different scales on the axis

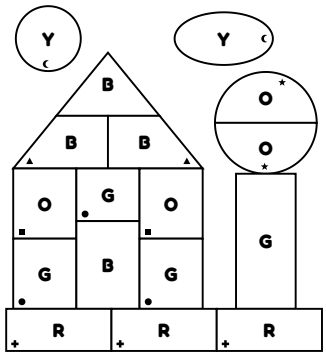
Syllabus outcomes: Bar Graphs 1.1, Bar Graphs 1.2. Notes on the tag: 1.1, 1.2 (reading and interpreting bar graphs; different scales on the axis)
Puzzle type: Code-breaker (two bar graphs). How the pupil checks: Code word (6 letters).
Answers: 1) 2; 2) 30; 3) 6; 4) 90; 5) 46; 6) 60
Check result: Code word: TUXEDO
Teacher note: Two bar graphs with different scales: Graph A goes up in 2s (a line for every 2, a number for every 4) and Graph B goes up in 10s (a line for every 10, a number for every 20). Each question names its graph. The decoy letters stand for reading Graph B as if its lines were 2 apart or Graph A as if its lines were 10 apart, and for the next line. The bars are black. A box left blank gives no letter and the check cannot see it.

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S2-19 (P3-39) Colour the House 2 stars

Topic: Length, mass and volume: km and m, m and cm, kg and g, ℓ and mℓ

Syllabus outcomes: Length, Mass and Volume 1.2, Length, Mass and Volume 1.3. Notes on the tag: 1.2, 1.3 (compound units; converting km and m, m and cm, kg and g, ℓ and mℓ)
Puzzle type: Colour by code (pupils colour with their own crayons). How the pupil checks: Colour key and tally: every answer must be on the key, the number of parts of each colour must match the box, and parts with the same small mark match.
Answers: Colour key: 1500 = yellow, 2500 = red, 3000 = blue, 4500 = green, 6000 = orange
Check result: Tally: yellow 2, red 3, blue 4, green 4, orange 4; parts with the same small mark match.

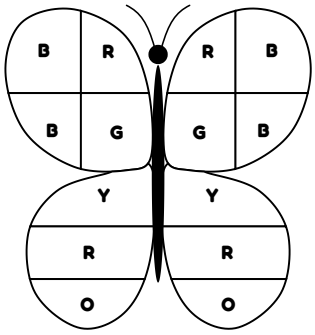


Teacher note: Every part changes units: km to m, m to cm, kg to g, ℓ to mℓ, or adds or takes away a part in the smaller unit. Pupils colour the black-and-white outline with their own crayons (or, for a pencil-only copy, write Y, R, B, G or O in each part: the tally and the matching parts can be checked the same way). Check strip: count the parts of each colour (the tally is printed); parts with the same small mark match.

S2-20 (P3-40) Colour the Butterfly 2 stars

Topic: Money: adding and subtracting in dollars and cents

Syllabus outcomes: Money 1.1. Notes on the tag: 1.1 (adding and subtracting money in decimal notation)
Puzzle type: Colour by code (pupils colour with their own crayons). How the pupil checks: Colour key and tally: every answer must be on the key, the number of parts of each colour must match the box, and the parts in the same place on the left and on the right match.
Answers: Colour key: $2.35 = yellow, $4.50 = red, $6.80 = blue, $8.05 = green, $12.75 = orange
Check result: Tally: yellow 2, red 4, blue 4, green 2, orange 2; the parts in the same place on the left and on the right match.

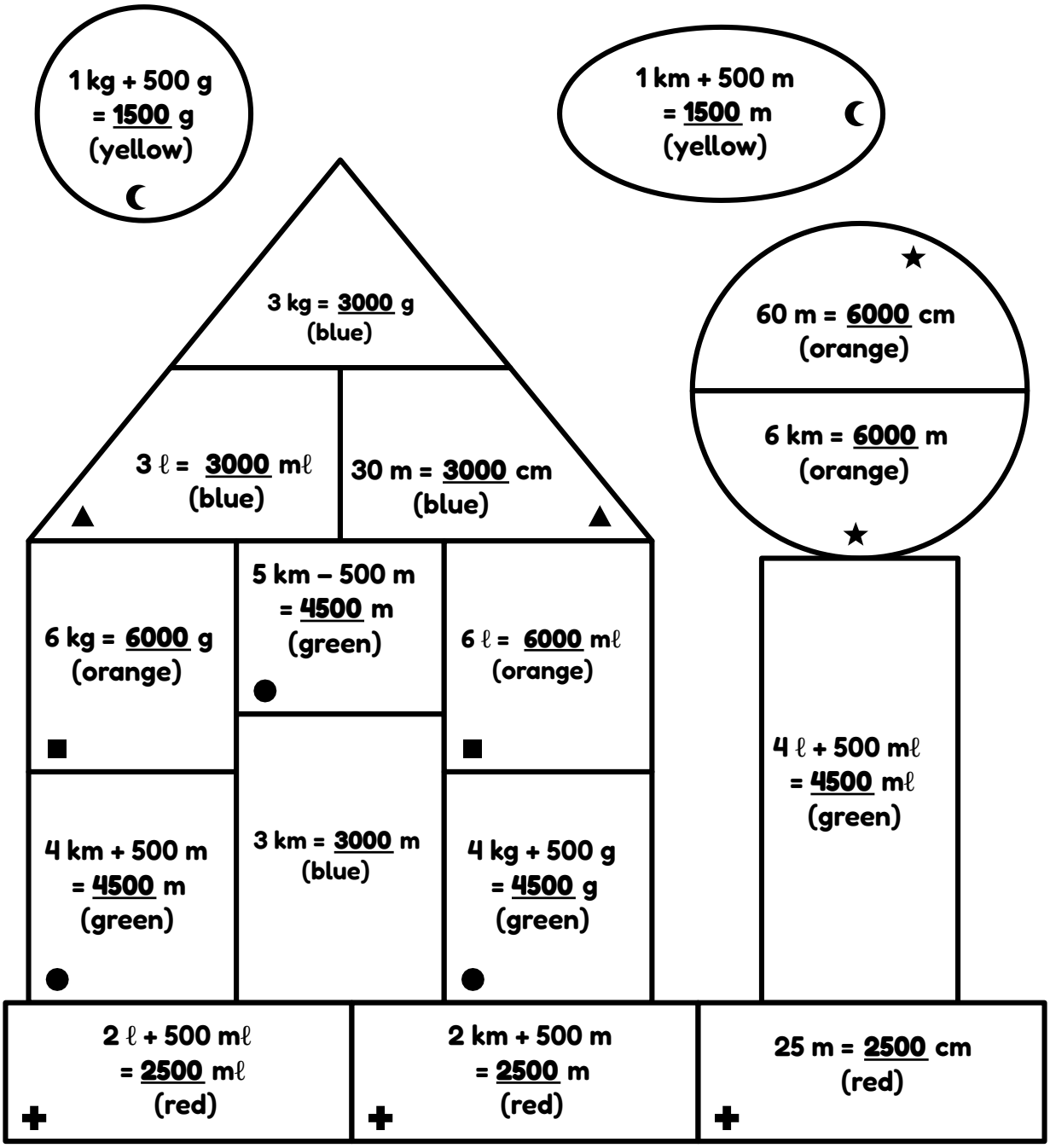


Teacher note: Every part adds or subtracts amounts of money in dollars and cents: two or three prices, money left, and change from a note. Pupils colour the black-and-white outline with their own crayons (or, for a pencil-only copy, write Y, R, B, G or O in each part: the tally and the matching parts can be checked the same way). Check strip: count the parts of each colour (the tally is printed); the parts in the same place on the left and on the right match.

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Answers: S2-19 (P3-39) Colour the House

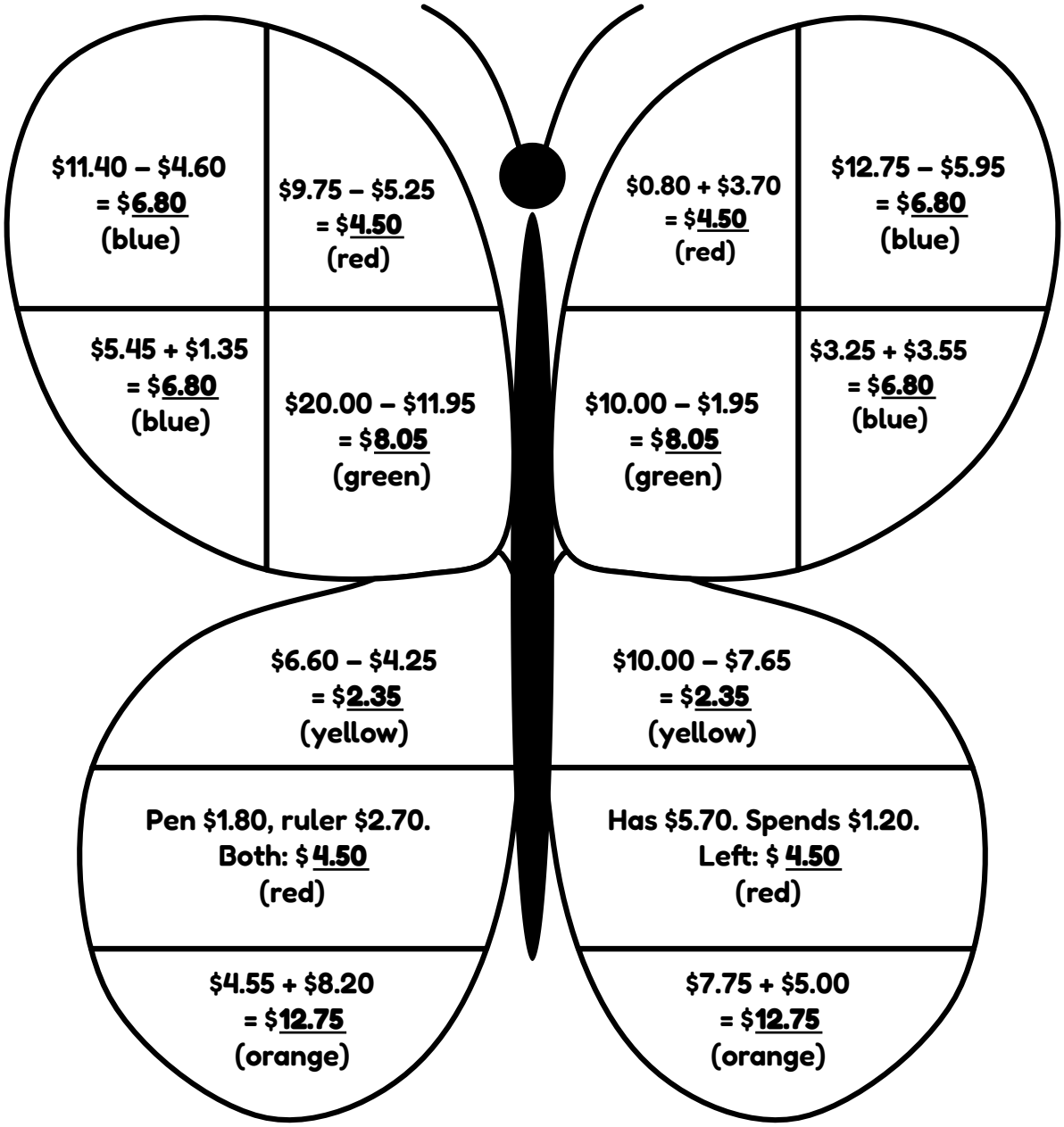
Every part shows the question, the answer underlined and the colour in brackets. Pupils colour their own copy. Colour key: 1500 = yellow, 2500 = red, 3000 = blue, 4500 = green, 6000 = orange.



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Answers: S2-20 (P3-40) Colour the Butterfly

Every part shows the question, the answer underlined and the colour in brackets. Pupils colour their own copy. Colour key: $2.35 = yellow, $4.50 = red, $6.80 = blue, $8.05 = green, $12.75 = orange.



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